

EKO S. WIBOWO

Freelance Full-Stack Programmer & Technical Trainer

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PROFESSIONAL PROFILE

Software developer since the late 1990s, currently active as a freelance full-stack programmer. I design, build, and operate complete application estates — from backend services and frontends to infrastructure provisioning and deployment — using my own Domain-Driven “Eco” framework (see *Current Work*), and I ship mobile apps in Flutter (Quran Guardian). Winner of the Facebook Developer Community Challenge, 1st place for the Asia Pacific region (2018). Former national consultant for the Food and Agriculture Organization of the United Nations (FAO UN). Experienced software engineering trainer and conference speaker (PyCon Indonesia 2019). ## LATEST AWARD

Winner — Facebook Developer Community Challenge, Asia Pacific Region (1st place) — Aug 2018

Built **Whizkids.id**, a gamified platform for teaching kids how to code, set in an isometric RPG open-world guided by a bot, teacher, and parents. Achieved **1st position in the Asia Pacific region** and **7th position worldwide** — my winning project out of a global field of entries, and a highlight of my career as a solo developer.

- Announcement: <https://devcommunitychallenge.devpost.com/updates/8710-the-regional-winners-are>
- Project: <https://devpost.com/software/whizkids-id-virtual-world>

Silver Award — Digital Start-Up Category, ASEAN Digital Awards 2024 — Artopologi (co-founder & CTO) ## CURRENT WORK — FREELANCE PROGRAMMER

Since 2024 I work as a freelance full-stack programmer, building a family of production systems on the **Eco framework** I created and continue to develop: a host-native Domain-Driven Design (DDD) platform that decomposes projects into independent, self-sustaining *domains* (auth, photos, chat, bidding, rag, ...) which are recomposed into new projects — *estates* — declared in a single `ecompose.yml` manifest. One Proxmox CT is the machine boundary; services run natively (no Docker) under PM2, exposed through Cloudflare Tunnel. Docker-like reproducibility (`eco up`), webhook deploys on push to main. Public documentation lives at <https://eco.stuff8.com>.

#1. eco — the framework itself — <https://eco.stuff8.com> Primary implementation of the Ecology DDD workflow as a private Node CLI: estate provisioning on Proxmox, runtime wiring, .env generation, PM2/Caddy orchestration, and Cloudflare tunnel exposure. Ships a bundled Bash workflow for workspace-side orchestration and Proxmox CT lifecycle commands (`eco ct`, `eco up`, `eco configure`, `eco expose`).

#2. stuff8 — personal inventory to public marketplace — <https://stuff8.com> Production estate built with Eco: ten independent domains (auth, photos, inventory, marketplace, bidding, chat, notifications, profile, rag, plus an Astro.js + Tailwind CSS frontend) composed into one application deployed on a single CT. “Inventory-first, marketplace-second”: a personal asset inventory whose items can be toggled sellable into a public marketplace with offers, negotiation via realtime chat, and COD transactions. Demonstrates the full composition model across Rust (axum), Go, and Node runtimes.

#3. assessment — psychometric assessment platform for Indonesian schools Rebuild of a legacy Laravel “ODAS+IST” system into a modern SvelteKit 2 frontend (port 2001) + Java Spring Boot 3 backend (port 2002) sharing a JWT auth server. Five psychometric instruments are implemented end-to-end: **DISC**, **Holland RIASEC**, **PAPI Kostick** (with radar-chart results), **IQ CFIT**, and **IQ IST** (image-based subtests scored server-side). Students sit standardized tests, get scored results, and download certificates; guidance counselors and affiliate psychologists administer assessments per school.

#4. training — online training / LMS platform An Eco estate composed of reusable domains — auth, community, content, courses, payments, profile, site, slides, plus an LMS frontend — rewritten in Rust (axum) from a legacy Java monolith. Handles course/community management, S3/local file uploads, payments, and landing-page/content delivery, deployed via the standard `ecompose.yml` + `eco up` flow.

#5. Quran Guardian — Flutter Quran reading app — <https://quranguardian.vercel.app> A Flutter mobile app for immersive Quran reading and habit-building: contribution-calendar-style streak/consistency tracker, Instagram-

like hadith & Islamic-history stories, distraction-free full-screen reading sanctuary, verse-by-verse synchronized audio (Mishary Alafasy) with offline downloads, and a geographical reading-journey map with prayer-time count-downs. Content pipeline combines **Python scraping** of **Quran.com** structured data (translations, tafsir, verse metadata, word-by-word) persisted in **SQLite** for fast offline access on-device. Shows my full mobile-app readiness — Flutter on Android, backed by a Python/scripting data layer.

#6. MicroCMS — enterprise digital signage platform — <https://cms.microvision.co.id> Multi-tenant digital signage CMS built with a **Next.js + Supabase** web backend, driving multiple presentation players: a **Java Android** app and a **.NET desktop** player. Supports real-time monitoring, over-the-air deployment/updating of content, Canva-like drag-and-drop layout management, and **LG WebOS** signage playback. Enterprise-grade security with admin-controlled access and role-based permissions (Owner / Admin / Member / Viewer). ## PROGRAMMING LANGUAGES & TECHNICAL SKILLS

Programming Languages

Category	Technology
Python (primary, 15+ yrs)	Flask, Pyramid, Django, Tornado, FastAPI, scikit-learn, TensorFlow, BeautifulSoup
Rust	axum, tokio, MongoDB/PostgreSQL drivers, fastembed/ONNX
Go	WebSocket hubs, MongoDB
Java	Spring Boot 3, Swing, JSP/Servlet, J2ME, JDBC
C / C++ / C# / .NET	C#.NET, VB.NET, VC++ .NET, Visual Basic 6, MIDI/Atmel tooling
PHP	CodeIgniter, pure PHP, WordPress, ExtJS
JavaScript / TypeScript	SvelteKit 2, Svelte 5, Astro.js, Node.js, Next.js, Dojo, jQuery
Dart / Flutter	Flutter (Android), custom widgets, on-device SQLite
SQL & NoSQL	PostgreSQL, MySQL, MSSQL, Oracle, SQLite, MongoDB, Redis, Supabase

Full-Stack on the Eco Framework

- **Frontend:** SvelteKit 2 + Svelte 5 + Vite, Astro.js + Tailwind CSS, shadcn-svelte; classic Dojo/ExtJS/jQuery when called for.
- **Backend:** REST APIs in Python (Flask/Django/Tornado/FastAPI), Rust (axum), Go, Java Spring Boot; server-sent scoring and test engines.
- **Infrastructure & Deployment:** Proxmox VE (CT lifecycle, pct, PM2), Docker, AWS, Heroku, Caddy gateway, Cloudflare Tunnel; ecompose.yml manifests + eco up provisioning; push-to-deploy webhook auto-redeploys.
- **Data & Messaging:** PostgreSQL, MongoDB, MySQL, MSSQL, Oracle, SQLite, Redis, RabbitMQ, MinIO/S3, Kafka.
- **OS:** Linux (Red Hat, Ubuntu), Windows (95/XP/7), macOS, plus VirtualBox virtualization.
- **Methodology:** Domain-Driven Design, Extreme Programming, Scrum (incl. Scrum Master), Kanban, code review & CI.

Other

- **Mac / iOS / Mobile:** Flutter (Android), macOS application packaging (Wine/Bottler), C# to Mac porting, Cocos2d & Cocos2d-x games, Core Data, Objective-C, .NET Compact Framework, J2ME mobile apps.
 - **Digital Signage & Embedded Players:** Java Android signage player, .NET desktop player, LG WebOS playback, Canva-like drag-and-drop layout editor, real-time device monitoring and OTA content deployment.
- ## CAREER PORTFOLIO & WORK HISTORY

Recent Freelance & Current Roles

#1. Freelance Full-Stack Programmer — 2011 – present Long-running freelance practice serving foreign companies through remote work, plus independent product and estate development. See *Current Work* for the eco, stuff8, assessment and training systems I currently build and operate.

#2. Co-Founder & CTO — Artopologi (artopologi.com) — ongoing Indonesian art marketplace with blockchain-backed certificate of authenticity. Assembled and led a fully remote team to launch; still in development and maintenance. Won **Silver in the Digital Start-Up Category at the ASEAN Digital Awards 2024**.

#3. National Consultant — FAO UN / Ministry of Forestry Indonesia — Feb 2012 – present Contracted by FAO headquarters (Rome) to develop **Open Foris Collect** — the National Forest Inventory (NFI) database application framework — to suit Indonesia-specific needs. One of the strongest public-sector references in my career.

Notable Software Roles

#4. MonetizeMore — Feb 2016 – Nov 2019 Google AdSense partner company optimizing high-revenue publisher websites (USD 20k+/month) via AdExchange/DFP and Header Bidding. Stack: Tornado, Python Flask, RabbitMQ, Docker, AWS, BeautifulSoup web scraping for automated publisher reports. Served as Scrum Master leading a national team; left on excellent terms (Employee of the Quarter, excellence performance reviews).

#5. Python Developer — ScramSoftware (Australia) — Aug 2014 – Oct 2015 Longest full-time remote contract; yearly engagement building **ScramGet** (auto-fetch of user data from email, YouTube, DropBox, Google Drive). Forked youtube-dl for company-specific needs and integrated it from a Kivy application.

#6. System Designer — PT. Gamatechno Indonesia — 2006 – 2007 Java web framework development (Web-work); left to start my own software consultation service.

#7. Java Game Developer — PT. Gameloft Indonesia — Jan – May 2011 J2ME/mobile game development, including mass-porting *Diamond Twister 2* across devices (Sony Ericsson K800 to Nokia 6280 to BlackBerry).

Selected Past Projects

- **SaleStock Indonesia** (Oct–Dec 2015) — A/B-test microservices in Python integrated with Scala backend.
- **PassBandara Management System** (Dec 2011–Feb 2012) — Airport pass digitalization, Ministry of Transportation, VB.NET + MySQL.
- **Fikr-Malaysia — Distributed POS** (Jun 2013–) — C#.NET desktop POS + Pyramid/PostgreSQL central API (Heroku).
- **Time.ly WordPress Calendar Plugins — Technical Support** (Apr–Aug 2013).
- **Python Technical Writer — pythonthusiast.com** (Oct 2013–May 2014) — web (Flask, Django), Panda3D, Cocos2d-x, Qt/PyQt/PySide.
- **SiGAP** (2000–2001) — accounting & inventory control for a Jepara furniture manufacturer, PT. Detro Consulting.
- **Merpati META** (2001–2002) — DSS predicting market share for PT. Merpati Nusantara Airlines (Ministry of Research; published in the East Asia Transportation Journal).
- **SWDEV Software Consulting** (2007–2011) — my own consultation service spanning .NET, J2ME, Java SMS gateways (20+ apps), PHP/CodeIgniter e-commerce, JSP e-education, and more.

Teaching & Early Career

- **Computer Programming Instructor** — Bugs Training Center & Bijak Training Services (2002–2005).
- **IT Lab Assistant** — UGM Student Internet Center, Computer Science faculty (2006–2007).
- **ExtremePOS** (2007) — first freelance job: CRM/ERP/POS for a mini-market chain. ## TRAINER & SEMINAR HISTORY

#0. Speaker — PyCon Indonesia 2019 (Surabaya) “How I (almost) Single Handedly Run a Community Platform using Odoo and TelegramBot”.

#1. Digital Fintech Workshop — PT. Berdiklat.com (Sep 2019, Jogjakarta) Two-day training for IT & Marketing staff of Bank Exim Indonesia — Fintech fundamentals and adoption strategy using Python and its infrastructure.

- #2. Trainer ECOMMERCE B2B — TELKOM** (Feb 2019, Jogjakarta) For Telkom Bandung Marketing Division — business case development and e-commerce with Python Django.
- #3. Data Science & Machine Learning Workshop — BPK-RI** (Dec 2018, Jakarta) Five-day training for IT staff of Badan Pengawas Keuangan RI, from Python fundamentals to scikit-learn and TensorFlow.
- #4. Odoo Python Training — Berdiklat.com** (Feb 2016, Jogjakarta) For IT staff at BPD DIY — maneuvering within Odoo's Python codebase.
- #5. JOGJAFAMA — Jogja Freelancer Ask Me Anything** (Sep 2015) — endorsed by Freelancer.com.
- #6. STMIK Jenderal A. Yani** (May 2016) — “Technopreneur-ready students competing in the global job market”.
- #7. Universitas Gadjah Mada** (Apr 2017) — “Global Freelancing and Digital Entrepreneurship”.
- #8. Universitas Pelita Bangsa, West Java** (Aug 2017).
- #9. Webinar — “Earning thousands of dollars a month as a freelancer”**, endorsed by Payoneer.
- #10. Skyshi Academy — Python Fundamentals & Architecture** (online discussion).
- #11. Python ID meetups** (Jogja, from Mar 2018) and **#12. Unisnu Jepara** seminar on remote work — plus many more campus & community talks across Indonesia. ## EDUCATIONAL RECORD

Time Period	Institution	Location	Major
2021 – Now	Universitas Islam Indonesia	Yogyakarta, Indonesia	Economics & Finance (Magister)
1998 – 2008	Gadjah Mada University	Yogyakarta, Indonesia	Computer Science
1995 – 1998	SMU Negeri 3	Denpasar, Bali, Indonesia	High School
1992 – 1995	SMP Negeri 5	Denpasar, Bali, Indonesia	Intermediate School
1992 – 1998	SD Negeri 5	Denpasar, Bali, Indonesia	Elementary School

PERSONAL IDENTITY

- **Place & Date of Birth:** Denpasar, May 27th 1979
- **Nationality:** Indonesia
- **Marital Status:** Married
- **Sex:** Male
- **Languages:** Indonesian and English (fluent in speaking)

Professionalism: Freelance Full-Stack Programmer, Software Developer, Computer Programming Instructor, and a decent Capoeira player.